

Arunabha Majumder

Mechanical Engineer — Arm-Worn Wearables, Prototyping & Fit Testing (SolidWorks)

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SUMMARY

Mechanical engineer for arm-worn devices. Designed and built two upper-limb wearables, prototyped them through 3D printing and tested them on real arms with participants. Also built soft pneumatic actuators by hand. SolidWorks (CSWA), Best Paper Award 2026.

KEY RESULTS

- 2 upper-limb wearables designed, built and tested on people
- >25% energy reduction from the wearable's actuation and mounting architecture
- >15% lower wearer muscle effort, measured with EMG

EXPERIENCE

Research Engineer – Wearable Arm & Shoulder Devices (PhD Project: VIEXO) | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Designed and prototyped a body-worn shoulder and upper-arm device in SolidWorks, iterating through 3D-printed prototypes and testing on real arms with 12 participants.
- Cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026) through the wearable's actuation and mounting architecture: a parallel spring working with a 6 Nm motor.
- Integrated sensors and electronics into the wearable (motor, encoder, force sensors, DAQ, cabling), working across mechanical, electronics and control.
- Measured human arm impedance with a force-estimation method on a 5-bar planar parallel robot, validated against a force sensor to below 10% RMS error.
- Reduced wearer muscle effort by more than 15%, measured with EMG, through Assist-as-Needed control; mentored Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Soft Pneumatic Actuators (Master's Thesis) | 2020 – 2022 *CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Built soft pneumatic artificial muscles from raw materials and their test rig (valves, pressure sensors, DAQ); reduced tracking error to 0.3–0.78% with a neural-network-based controller.

Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton (Medical Wearable) | 2018 – 2019 *Siddaganga Institute of Technology, Tumakuru, India*

- Won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant by leading CAD design, machining, hands-on assembly and user testing.

SKILLS

- **Wearables:** Arm and shoulder exoskeletons, body-worn mechanisms, testing with participants
- **CAD & prototyping:** SolidWorks (CSWA), 3D printing (FDM), rapid iteration; machining and assembly (Bachelor's)
- **Actuators & sensors:** Soft pneumatic artificial muscles, motors, encoders, force sensors, DAQ
- **Software:** MATLAB/Simulink, Python, Arduino
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present *Aalborg University, Department of Materials and Production, Aalborg, Denmark*

M.Tech, Mechatronics | 2020 – 2022 *AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

REFERENCES

Available upon request.